databricks Set-up / 6.mount_adls_container_for_project

(https://databricks.com)

```
def mount_adls(storage_account_name, container_name):
    # Get secrets from Key Vault
    client_id = dbutils.secrets.get(scope = 'formula1-scope', key = 'formula1-app-client-id')
    tenant_id = dbutils.secrets.get(scope = 'formula1-scope', key = 'formula1-app-tenant-id')
    client_secret = dbutils.secrets.get(scope = 'formula1-scope', key = 'formula1-app-client-secret')

    # Set spark configurations
    configs = {"fs.azure.account.auth.type": "OAuth",
          "fs.azure.account.oauth.provider.type": "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider",
          "fs.azure.account.oauth2.client.id": client_id,
          "fs.azure.account.oauth2.client.secret": client_secret,
          "fs.azure.account.oauth2.client.endpoint": f"https://login.microsoftonline.com/{tenant_id}/oauth2/token"}

    # Unmount the mount point if it already exists
    if any(mount.mountPoint == f"/mnt/{storage_account_name}/{container_name}" for mount in dbutils.fs.mounts()):
        dbutils.fs.unmount(f"/mnt/{storage_account_name}/{container_name}")

    # Mount the storage account container
    dbutils.fs.mount(
      source = f"abfss://{container_name}@{storage_account_name}.dfs.core.windows.net/",
      mount_point = f"/mnt/{storage_account_name}/{container_name}",
      extra_configs = configs)

    display(dbutils.fs.mounts())
```

```
##### Mount Raw Container

mount_adls('formula1dl1204', 'raw')
```

Table

	mountPoint	source	encryptionType
1	/databricks-datasets	databricks-datasets	
2	/mnt/formula1dl204/demo	abfss://demo@formula1dl1204.dfs.core.windows.n...	
3	/Volumes	UnityCatalogVolumes	
4	/mnt/formula1dl1204/raw	abfss://raw@formula1dl1204.dfs.core.windows.net/	
5	/databricks/mlflow-tracki...	databricks/mlflow-tracking	
6	/databricks-results	databricks-results	
7	/databricks/mlflow-registry	databricks/mlflow-registry	
8	/Volume	DbfsReserved	
9	/volumes	DbfsReserved	
10	/	DatabricksRoot	
11	/volume	DbfsReserved	

11 rows

```
##### Mount processed Container

mount_adls('formula1dl1204', 'processed')
```

Table

	mountPoint	source	encryptionType

1	/databricks-datasets	databricks-datasets	
2	/mnt/formula1dl204/demo	abfss://demo@formula1dl1204.dfs.core.windows.net/	
3	/mnt/formula1dl1204/process	abfss://processed@formula1dl1204.dfs.core.windows.n	
4	/Volumes	UnityCatalogVolumes	
5	/mnt/formula1dl1204/raw	abfss://raw@formula1dl1204.dfs.core.windows.net/	
6	/databricks/mlflow-tracking	databricks/mlflow-tracking	
7	/databricks-results	databricks-results	
8	/databricks/mlflow-registry	databricks/mlflow-registry	
9	/Volume	DbfsReserved	
10	/volumes	DbfsReserved	
11	/	DatabricksRoot	
12	/volume	DbfsReserved	

12 rows

```
##### Mount presentation Container
mount_adls('formula1dl1204', 'presentation')
```

Table

	mountPoint	source	encryptionType
1	/databricks-datasets	databricks-datasets	
2	/mnt/formula1dl1204/presentati...	abfss://presentation@formula1dl1204.dfs.core.windows.n...	
3	/mnt/formula1dl204/demo	abfss://demo@formula1dl1204.dfs.core.windows.net/	
4	/mnt/formula1dl1204/processed	abfss://processed@formula1dl1204.dfs.core.windows.net/	
5	/Volumes	UnityCatalogVolumes	
6	/mnt/formula1dl1204/raw	abfss://raw@formula1dl1204.dfs.core.windows.net/	
7	/databricks/mlflow-tracking	databricks/mlflow-tracking	
8	/databricks-results	databricks-results	
9	/databricks/mlflow-registry	databricks/mlflow-registry	
10	/Volume	DbfsReserved	
11	/volumes	DbfsReserved	
12	/	DatabricksRoot	
13	/volume	DbfsReserved	

13 rows

```
client_id = dbutils.secrets.get(scope = 'formula1-scope', key = 'formula1-app-client-id')
tenant_id = dbutils.secrets.get(scope = 'formula1-scope', key = 'formula1-app-tenant-id')
client_secret = dbutils.secrets.get(scope = 'formula1-scope', key = 'formula1-app-client-secret')
```

```
configs = {"fs.azure.account.auth.type": "OAuth",
          "fs.azure.account.oauth.provider.type": "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider",
          "fs.azure.account.oauth2.client.id": client_id,
          "fs.azure.account.oauth2.client.secret": client_secret,
          "fs.azure.account.oauth2.client.endpoint": f"https://login.microsoftonline.com/{tenant_id}/oauth2/token"}
```

```
dbutils.fs.mount(
  source = "abfss://demo@formula1dl1204.dfs.core.windows.net/",
  mount_point = "/mnt/formula1dl204/demo",
  extra_configs = configs)
```

```
True
```

```
display(dbutils.fs.mounts())
```

Table

	mountPoint	source	encryptionType
1	/databricks-datasets	databricks-datasets	
2	/mnt/formula1dl204/demo	abfss://demo@formula1dl1204.dfs.core.windows.n...	
3	/Volumes	UnityCatalogVolumes	
4	/databricks/mlflow-tracki...	databricks/mlflow-tracking	
5	/databricks-results	databricks-results	
6	/databricks/mlflow-registry	databricks/mlflow-registry	
7	/Volume	DbfsReserved	
8	/volumes	DbfsReserved	
9	/	DatabricksRoot	
10	/volume	DbfsReserved	

10 rows

```
display(dbutils.fs.ls("/mnt"))
```

Table

	path	name	size	modificationTime
1	dbfs:/mnt/formula1dl/	formula1dl/	0	1721488475000
2	dbfs:/mnt/formula1dl20...	formula1dl204/	0	1721488522000

2 rows

```
display(dbutils.fs.ls("/mnt/formula1dl204/demo"))
```

Table

	path	name	size	modificationTime
1	dbfs:/mnt/formula1dl204/demo/circuits.c...	circuits.csv	10044	1721470200000

1 row

```
display(spark.read.csv("/mnt/formula1dl204/demo/circuits.csv"))
```

Table

	_c0	_c1	_c2	_c3	_c4	_c5	_c6	_c7
1	circuitId	circuitRef	name	location	country	lat	lng	alt
2	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	144.968	10
3	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	101.738	18

4	3	bahrain	Bahrain International Circuit	Sakhir	Bahrain	26.0325	50.5106	7
5	4	catalunya	Circuit de Barcelona-Catalunya	Montmeló	Spain	41.57	2.26111	109
6	5	istanbul	Istanbul Park	Istanbul	Turkey	40.9517	29.405	130
7	6	monaco	Circuit de Monaco	Monte-Carlo	Monaco	43.7347	7.42056	7
8	7	villeneuve	Circuit Gilles Villeneuve	Montreal	Canada	45.5	-73.5228	13
9	8	magny_cours	Circuit de Nevers Magny-Cours	Magny Cours	France	46.8642	3.16361	228
10	9	silverstone	Silverstone Circuit	Silverstone	UK	52.0786	-1.01694	153
11	10	hockenheimri...	Hockenheimring	Hockenheim	Germany	49.3278	8.56583	103
12	11	hungaroring	Hungaroring	Budapest	Hungary	47.5789	19.2486	264
13	12	valencia	Valencia Street Circuit	Valencia	Spain	39.4589	-0.331667	4
14	13	spa	Circuit de Spa-Francorchamps	Spa	Belgium	50.4372	5.97139	401

78 rows

```
display(dbutils.fs.ls("abfss://demo@formula1dl1204.dfs.core.windows.net"))
```

Table

	path	name	size	modificationTime
1	abfss://demo@formula1dl1204.dfs.core.windows.net/circuits.c...	circuits.csv	10044	1721470200000

1 row

Table

	_c0	_c1	_c2	_c3	_c4	_c5	_c6	_c7
1	circuitId	circuitRef	name	location	country	lat	lng	alt
2	1	albert_park	Albert Park Grand Prix Circuit	Melbourne	Australia	-37.8497	144.968	10
3	2	sepang	Sepang International Circuit	Kuala Lumpur	Malaysia	2.76083	101.738	18
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78 rows